What Does ODRL Mean? A Cross-Level Ontological Grounding of Permissions, Prohibitions, and Duties in UFO-L

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Abstract. ODRL policy evaluators produce verdicts, but say nothing about the normative positions a policy brings into existence, the authority structures those positions presuppose, or who holds the power to declare a norm violated. We formulate the Cross-Level Design Principle: any normative language with violable, consequential norms requires both conduct-level positions (Permission, Duty, Right, No right) and competence-level positions (Power, Subjection, Immunity, Disability). Applying this to ODRL, we establish that prohibition is *sanctioned* (violation possible and consequential), that permission is underspecified across its `behaviour` parameter (open vs. closed world), and that the formal semantics covers achievement obligations only. We ground ODRL in UFO-L, mapping each activated rule to a simple legal relator and extending coverage from two to eight legal positions; violation-declaration authority, implicit in every existing evaluator, becomes an explicit Power, Subjection pair. All axioms and key entailments are mechanically verified in Isabelle/HOL and across a 39-problem benchmark under Vampire, E, and Z3.

Keywords. ODRL, UFO-L, Foundational Ontology, Legal Positions, Normative Systems, Deontic Logic, Data Spaces

1. Introduction

The Open Digital Rights Language (ODRL) [1] is a policy language standardised by W3C.[2] It has been adopted by the European Data Spaces [2,3,4] as the lingua franca for expressing data usage and data access policies over shared assets. ODRL provides both an information model and an ontology [1] specifying how policies, rules, and constraints are structured. The Formal Semantics specifies the behaviour of an *ODRL Evaluator*, a software component that determines, given a set of policies and a state of the world, which rules are active, which actions are permitted or prohibited, and which obli-

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[2]Benchmark available at `https://github.com/Daham-Mustaf/odrl-ufol-grounding`

gations have been fulfilled or violated [5,6]. The evaluator's output is a verdict: `active`, `violated`, `fulfilled`, or `not-set` [5].
Consider `:KulturPortal`, an events portal that obtains the same theater showtime records for a cultural event from two Data Space providers, the theaters `:BerlinerEnsemble` and `:DeutschesTheater`, via separate connectors and contracts. Each issues an `odrl:Agreement` prohibiting redistribution of its showtimes, but with a different remedy: `:BerlinerEnsemble` demands compensation, `:DeutschesTheater` demands deletion of the copy.

```
:BE-Agreement a odrl:Agreement ;
 odrl:prohibition [
  odrl:assignee :KulturPortal ;
  odrl:assigner :BerlinerEnsemble ;
  odrl:action   odrl:distribute ;
  odrl:target   :BE-Showtimes ;
  odrl:remedy [      # if violated:
   odrl:action
    odrl:compensate ;# pay theater
   odrl:assignee
    :BerlinerEnsemble ] ] .
```

```
:DT-Agreement a odrl:Agreement ;
 odrl:prohibition [
  odrl:assignee :KulturPortal ;
  odrl:assigner :DeutschesTheater ;
  odrl:action   odrl:distribute ;
  odrl:target   :DT-Showtimes ;
  odrl:remedy [      # if violated:
   odrl:action
    odrl:delete ;    # destroy copy
   odrl:assignee
    :KulturPortal ] ] .
```

`:KulturPortal` distributes both datasets. The ODRL Evaluator reports `violated` for both policies, but cannot determine by whose normative authority either remedy is enforceable: it sees the remedy assignees but has no account of which provider holds the Power to declare a violation and demand remedy performance. Neither prohibition carries a constraint, so both are unconditionally active; since the regulated action was performed, both reach `violated` [5]. Each prohibition specifies a remedy, but the Formal Semantics defines no evaluation procedure for remedy duties and the Evaluation Report remains unspecified [5]. The evaluator goes no further: it cannot determine (*i*) by what normative standing either provider may demand remedy performance, or (*ii*) what normative standing grounds either remedy, because the normative positions that would license these answers (Figure 1, Row 6), each provider's Right to Omission and their Power to declare a violation, are absent from both policy graphs.

What normative positions exist when an ODRL policy is in force, and what authority structures do those positions presuppose? The absence of an answer has four concrete manifestations: **(P1)** the `behaviour` parameter selects between evaluation regimes that create ontologically incompatible normative positions, with no account of which positions exist under each setting; **(P2)** violation-declaration authority is absent from the policy graph and silently delegated to implementors, causing incompatible runtime determinations across deployments; **(P3)** every prohibition entails a Right to demand omission for the assigner, yet neither evaluator nor policy graph represents it, leaving remedy selection without a normative basis; and **(P4)** no existing vocabulary enables governance frameworks to declare sanctioning authority at policy-authoring time.

We use UFO-L as a foundational ontology to determine what kinds of things exist when a policy is in force, not to give ODRL a formal-semantics denotation. We then *ground* ODRL in UFO-L [7,8] to address these problems: P1 by ontologically characterising both `behaviour` regimes; P2 and P4 by constituting a Power–Subjection pair at activation; and P3 by surfacing the assigner's Right to Omission as an entailed correlative, extending evaluator coverage from two to eight legal positions.

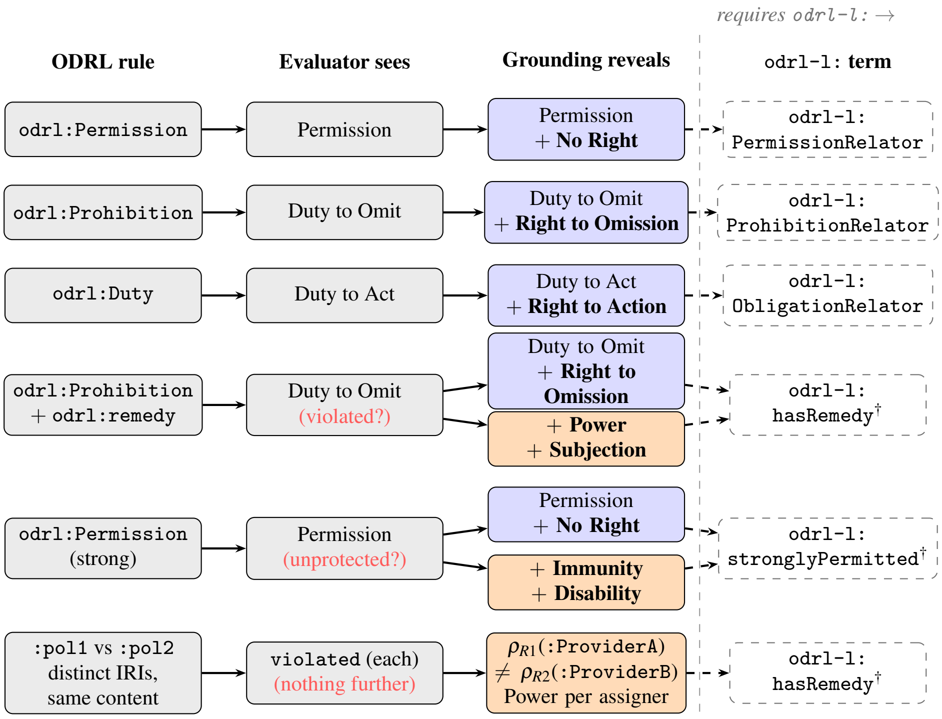

Figure 1. From an ODRL rule to its UFO-L grounding and the `odrl-l:` profile term. Each row reads left to right: the ODRL rule, what a native ODRL evaluator computes, the full simple legal relator the grounding reveals, and the `odrl-l:` term needed to represent it (Table 1). Shading: native ODRL; conduct - and competence -level positions; dashed outlines mark the `odrl-l:` profile. **Bold** marks positions absent from the evaluator's output; (?) marks where the evaluator halts; † marks the two ODRL extension points. A prohibition with a remedy (row 4) and a strong permission (row 5) each found a conduct relator and a competence relator at one activation event. Row 6 instantiates the motivating scenario of Section 1: distinct remedy relators per assigner make violation-declaration authority provider-specific.

2. Background

2.1. Open Digital Rights Language ODRL

An ODRL policy $\pi = \langle P, F, D\rangle$ consists of three disjoint rule sets: `odrl:Permission` rules P, `odrl:Prohibition` rules F, and `odrl:Duty` rules D. Each rule $r = \langle a, x, y, t, C\rangle$ specifies an action type a, an assignee x, an assigner y, a target t, and a constraint set C [1]. The action a is a *type* of act rather than a token, since granting a normative position over a single past token event would be ontologically incoherent: a policy regulates a kind of act, not one completed occurrence of it. The target t, by contrast, is a *particular* resource (a token), while the assignee x and assigner y may be either particular agents (tokens, as in an `odrl:Agreement`) or role types (as in an `odrl:Offer`). ODRL defines three `odrl:Policy` subclasses by the degree to which parties are identified [1]. An `odrl:Agreement` binds two identified particular agents, creating a bilateral normative bond. An `odrl:Offer` is binding *erga omnes* (toward all): the assigner is identified while the assignee is the community satisfying the offer's conditions [9]; it must not be evaluated by an ODRL Evaluator [5]. An `odrl:Set` is type-level: it creates no binding bond and falls outside the scope of this grounding [1]. `odrl:Permission` allows an action when all constraints are satisfied. `odrl:Prohibition` disallows an action but remains violable: the prohibited action is performable and its performance triggers `violated` [5]. `odrl:Duty` obliges an action: `fulfilled` upon performance and `violated` when performance is no longer possible [5]. Duties appear in four structural roles: policy-level obligations (`odrl:obligation`), permission conditions (`odrl:duty`), prohibition remedies (`odrl:remedy`), and failure consequences (`odrl:consequence`) [1]. The remedy and consequence roles form a cascade defined in the information model [1]: if the prohibition is infringed, all remedy Duties MUST be fulfilled; if a remedy Duty is not fulfilled, its consequence Duties become active. The Formal Semantics defines no activation procedure for either [5]. When a Permission and Prohibition co-apply, the `odrl:conflict` property selects a resolution strategy: `perm`, `prohibit`, or `invalid` (default) [1]. An ODRL rule *prescribes*; a legal position *exists*. `odrl:Permission`, `odrl:Prohibition`, and `odrl:Duty` are policy-graph nodes, authored by a policy writer, evaluated by an ODRL Evaluator. The legal positions these rules generate are ontological individuals that inhere in the parties at activation: they are not verdicts but normative facts about who may, must, or can do what, and in relation to whom.

The Formal Semantics defines `behaviour` as an optional evaluator input [5] governing how unmentioned actions are treated: under `open`, silence implies permission; under `closed` (the default), silence implies prohibition. Its ontological significance is that these two settings create incompatible normative positions.

ODRL Profiles [1] extend the vocabulary with new terms (e.g. Rule subclasses) defined on top of the Core Vocabulary; a Policy names the Profile(s) it uses via `profile`.

2.2. UFO-L: Foundational Ontology of Legal Positions

UFO-L [7,8] extends the Unified Foundational Ontology [10] with a formal theory of legal concepts based on Alexy's account of legal positions [11]. A *moment* is an existentially dependent particular inhering in exactly one bearer; by the non-migration principle [10, a67] it cannot migrate between bearers. An *externally dependent moment* inheres in one entity while remaining existentially dependent on a mereologically disjoint entity. A *relator* is a mereological aggregate of externally dependent moments [9]: each constituent moment inheres in its own bearer while depending on the counterparty, and the relator is their mereological sum, capturing the bilateral normative bond as a single ontological individual, founded by a unique event [10, a77]. *Mediation* is a derived notion, not a primitive: a relator mediates two entities precisely when it has parts inhering in each of them [12, p. 240].

Legal positions are moments borne by agents, classified as *Legal Entitlements* (positions implying an advantage) or *Legal Burdens/Lacks* (positions implying a constraint) [9]. Positions are always paired as correlatives within a *simple legal relator* bundling exactly one such pair [9]: at *conduct level* (governing what agents may, must, or must not do) the four types are Permission–No Right and Right–Duty; at *competence level* (governing the authority to bring normative positions into or out of existence through institutional acts [11,9]) they are Power–Subjection and Immunity–Disability. Their mapping to ODRL is shown in Table 1. Disability denies Power, rendering institutional acts void [8].

Immunity can arise from any source that constitutionally entrenches a position against revocation [8]. Legal relators are always brought into existence by a competent agent's Power [8], motivating the cross-level design principle.

3. Deontic Analysis of ODRL: Permission, Prohibition, and Duty

Permission, Prohibition, and Duty are the only ODRL constructs that generate normative positions between parties; Asset and Party are structural *relata*, and Constraint an *activation condition* that determines when a rule applies but creates no legal position [1].

3.1. Permission: Underspecified Between Weak and Strong

Strong permission persists under policy modification and requires an Immunity blocking any subsequent prohibition, a competence-level position [8,11]. In UFO-L terms, Permission to Act and Duty to Omit are *opposites*: they cannot be co-borne by the same agent over the same action and target. Under `behaviour=open`, undeclared actions carry no positive normative position: the evaluator's default is a convention, not a granted Permission. Under `behaviour=closed`, each declared permission creates a Permission–No Right relator. However, the Formal Semantics defines no persistence protection [5]: closed-world evaluation is consistent with but not equivalent to strong permission. The `behaviour` parameter therefore selects between ontologically incompatible normative positions, a distinction invisible to the evaluator but explicit in the UFO-L account (P1).

3.2. Prohibition: Provably Sanctioned

Prohibitions are either *regimented*, violation structurally impossible or *sanctioned*, violation possible but consequential.

Proposition 3.1. ODRL prohibition is *sanctioned*: the prohibited action remains performable and its performance triggers normative consequences.

Proof. Assume regimentation. Then [1], 'if the Prohibition has been *infringed by the action being exercised*, then all remedies MUST be fulfilled", describes an unreachable scenario, rendering `odrl:remedy` permanently inactivatable. Yet the Recommendation defines, exemplifies, and requires implementors to support it [1]. Contradiction. The Formal Semantics confirms independently via an explicit `violated` state [5]. Full proof in Appendix A. Since prohibition is sanctioned and violation triggers a reparative duty, any complete grounding requires a competence-level Power–Subjection pair to ground the violation-to-remedy transition (P2). □

3.3. Duty: Achievement Semantics Only

Obligations divide into three types by their satisfier [13,14]: *achievement* (satisfied by a bounded event), *maintenance* (satisfied by an ongoing state), and *process* (satisfied by a continuous activity). Events cannot be ongoing, since being ongoing is a temporary property incompatible with the timeless character of events [14].

Proposition 3.2. The ODRL Formal Semantics covers achievement obligations only; maintenance and process obligations lack evaluation semantics.

Proof sketch. The three-state semantics, `fulfilled` upon a single performance event, `violated` when performance is no longer possible, `not-set` otherwise [5], presupposes an event-bounded satisfier. Maintenance and process obligations have no such satisfier: the evaluator defines no transition for a persisting state or an ongoing activity. Full proof in Appendix A. □

4. Cross-Level Design Principle

Any complete ontological grounding of ODRL requires competence-level legal positions. All results are conditional on three axioms formalising UFO-L [8] (FOL encoding in Appendix A): **A1** normative state changes require a triggering institutional event; **A2** such events require a competent agent; **A3** competence is a Power–Subjection pair. These axioms hold under both regulative and constitutive readings of norm activation: even constitutive rules presuppose a prior creation event grounded in an agent's Power [8].

Definition 4.1 (Level-Complete and Cross-Level)**.** Let N be a normative concept and L a level of legal positions (either conduct or competence). A set of legal positions $\mathscr{H}$ is an *adequate characterization* of N relative to axiom set $\mathscr{A}$ if $\mathscr{H}$ entails all normative consequences of N under any model extension consistent with $\mathscr{A}$. A *normative consequence* of N is a legal position entailed by N; quantifying over model extensions makes adequacy robust, requiring $\mathscr{H}$ to keep entailing those positions as further rules are added, not only in one fixed model. N is *level-complete* at L if it has an adequate characterization using only positions from L. N is *cross-level* if every adequate characterization requires positions from both conduct and competence levels.

Proposition 4.1 (Weak Permission is Conduct-Level-Complete)**.** Relative to **A1–A3**, weak permission for an assignee x to perform an action a on a target t, granted by an assigner y, is adequately characterised at the conduct level by $\{\mathsf{Permission}(x,a,t), \mathsf{NoRight}(y,a,t)\}$ (here x bears a Permission to Act and y the correlative No Right over a on t): no competence-level positions are required. This establishes that the cross-level requirement is non-trivial: only violable and consequential concepts require competence-level positions. Proof in Appendix A.

Theorem 4.1 (Strong Permission is Cross-Level)**.** Relative to **A1–A3**, strong permission of x to perform a on t, granted by y, has no adequate characterization using conduct-level positions alone. (Proof: Appendix A.)

Theorem 4.2 (Sanctioned Prohibition is Cross-Level)**.** Relative to **A1–A3**, sanctioned prohibition has no adequate characterization using conduct-level positions alone. (Proof: Appendix A.) A norm is *violable* if the regulated action remains performable despite the norm; it is *consequential* if violation activates a reparative normative position.

Theorem 4.3 (Violable Norms Require Both Levels)**.** Relative to **A1–A3**, any norm that is (i) violable and (ii) consequential has no adequate characterization using conduct-level positions alone. (Proof: Appendix A.)

Principle 1 (Cross-Level Design)**.** Any normative language with violable, consequential norms must provide competence-level positions (Power, Subjection, Immunity, Disability) alongside conduct-level ones (Permission, Duty, Right, No right): a conduct-only vocabulary cannot ground violation-to-consequence transitions regardless of implementation. (Proof: Appendix A.)

Since ODRL prohibition is sanctioned (Proposition 3.1) and remedies are normative consequences of violation by definition [1], Theorem 4.3 entails that any complete grounding of ODRL requires competence-level positions. The Berliner Ensemble prohibition of Section 1 is exactly such a norm: its remedy is enforceable only if some party holds the Power to declare the violation, a competence-level position that no conduct-only vocabulary supplies (grounded concretely in Example 5.1).

5. Formal Grounding of ODRL in UFO-L

Each activated ODRL rule maps to a *simple legal relator* (Section 2.2), the minimal adequate grounding required by Theorem 4.3: a single rule creates exactly one correlative pair of legal positions, and a simple legal relator bundles exactly one such pair [8]. A *forbearance* $\mathit{rfr}(a)$ is the *omission* of action a: in UFO-L, refraining from performing a is a perdurant of kind omission, distinct from act a [7]. Accordingly, a Duty to refrain from a is a Duty to Omit a, and its correlative is a Right to Omission of a. We use $\mathsf{Forbearance}$ to denote the sort of all omissions; *Act* and $\mathsf{Forbearance}$ are disjoint sorts with $\mathit{rfr}(a) \in \mathsf{Forbearance}$ for every $a \in \mathit{Act}$, and the map rfr is injective. We write $\mathit{decl}(a)$ for the institutional act of declaring a violation of a; *decl* maps *Act* to *Act* injectively. Throughout, the eight legal positions are written in a three-place form: $\mathsf{Permission}(x,a,t)$ reads "x bears a Permission to Act over action a on target t", and likewise for Duty, Right, NoRight, Power, Subj, Immunity, and Disability.
Each active rule becomes a two-party legal relationship: one position for the assignee and the matching correlative for the assigner.

Definition 5.1 (Rules as Simple Legal Relators)**.**

$$\begin{aligned}
\textbf{Permission } \rho_P &: \ \mathsf{Permission}(x,a,t) \wedge \mathsf{NoRight}(y,a,t), \quad a \in \mathit{Act} \\
\textbf{Strong permission } \rho_I &: \ \mathsf{Immunity}(x,a,t) \wedge \mathsf{Disability}(y,a,t) \ \text{[only if } p \text{ is strongly permitted]} \\
\textbf{Prohibition } \rho_F &: \ \mathsf{Duty}(x,\mathit{rfr}(a),t) \wedge \mathsf{Right}(y,\mathit{rfr}(a),t), \quad \mathit{rfr}(a) \in \mathsf{Forbearance} \\
\textbf{Prohibition remedy } \rho_R &: \ \mathsf{Power}(y,\mathit{decl}(a),t) \wedge \mathsf{Subj}(x,\mathit{decl}(a),t) \ \text{[only if } f \text{ has } \texttt{odrl:remedy}\text{]} \\
\textbf{Duty } \rho_D &: \ \mathsf{Duty}(x,a,t) \wedge \mathsf{Right}(y,a,t), \quad a \in \mathit{Act}
\end{aligned}$$

Each rule founds exactly one relator via *founds* (the conduct relators ρ_P, ρ_F, ρ_D); additionally, a prohibition with `odrl:remedy` founds ρ_R via *founds_rem*, and a strong permission founds ρ_I via *founds_imm*, both at the same activation event. The distinct predicates keep the two relators separate.

Table 1. ODRL rules, UFO-L relators, and legal positions surfaced at activation [8]. • correlative; ■ added by grounding; † profile extension (`odrl-l:`); EE = assignee; ER = assigner.

ODRL rule	Relator	Position	Bearer	Level
`odrl:Perm.` (weak)	ρ_P	Permission to Act	EE	conduct
		No Right •	ER	conduct
`odrl:Perm.` (strong†)	$+\rho_I$	Immunity ■†	EE	competence
		Disability ■†	ER	competence
`odrl:Prohibition`	ρ_F	Duty to Omit	EE	conduct
		Right to Omission •	ER	conduct
`odrl:Proh.+remedy`†	$+\rho_R$	Power ■†	ER	competence
		Subjection ■†	EE	competence
`odrl:Duty`	ρ_D	Duty to Act	EE	conduct
		Right to Action •	ER	conduct

Strong perm. founds ρ_P (weak row) *and* ρ_I at same event (Ax5.2). †`odrl-l:hasRemedy` (Ax5.4); `odrl-l:stronglyPermitted` (Ax5.2). Immunity not exclusive to strong permission [8]. Founding: ρ_P, ρ_F, ρ_D via *founds*; ρ_I via *founds_imm*; ρ_R via *founds_rem*.

5.1. Resolving Underspecifications

Proposition 5.1 (Open-World Normative Vacuum)**.** Under `behaviour=open`, for any action a not regulated by a rule in π, the position set $Pos(\pi, e)$ (Definition 5.2) contains no Permission, No Right, Duty, or Right over $\langle a, t\rangle$ for any bearer. (Proof: Appendix A.)

P1: Behaviour parameter. Proposition 5.1 establishes that under `open` the open-world default is a runtime convention with no ontological counterpart in $\mathscr{A}$. Under `closed`, each declared permission creates relator ρ_P; adding $\mathsf{Immunity}(x,a,t) \wedge \mathsf{Disability}(y,a,t)$ realises strong permission persisting under subsequent normative change (Theorem 4.1). The `behaviour` parameter is thus a choice between ontologically incompatible normative positions, not merely a configuration switch.

P2: Violation authority. For every prohibition with a remedy, the grounding adds $\mathsf{Power}(y, decl(a), t) \wedge \mathsf{Subj}(x, decl(a), t)$, surfacing the evaluator's implicit role as normative authority. The Power constituted at activation licenses y not only to declare the violation directly but also to *delegate* sanctioning competence to a third party, creating a new Power–Subjection pair in another agent via a normative transfer act [8]. Governance frameworks can thereby specify *which* party holds violation-declaration authority for *which* prohibitions (Theorem 4.2).

P3: Assigner correlatives. The grounding surfaces a Right to Omission for every assigner of a prohibition and a Right to an Action for every assigner of a duty (Table 1, Axiom 5.3). These correlatives are ontologically entailed but absent from the ODRL evaluator: without them, no reasoner can identify whose normative standing grounds a given remedy, even when dataset identity is known.

P4: Normative authority. The Power–Subjection pair constituted at activation (Axiom 5.4) makes violation-declaration authority explicit, enabling data sovereignty representation in governance frameworks.

5.2. First-Order Axiomatization

Each axiom answers the same question: when an ODRL rule activates, what legal individuals come into existence? The answer is always a simple legal relator that bundles one correlative pair, one position per party, founded by the activation event and aggregated via *partOf*. *Predicate key. ODRL rule (policy graph):* $\mathsf{P}(p)$/$\mathsf{Proh}(f)$/$obl(d)$ = Permission/Prohibition/Duty rule node (*obl*, the Duty *rule*, is distinct from Duty, the UFO-L Duty *position*); $aee(r,x)$/$aer(r,y)$ = r has assignee x / assigner y; $act(r,a)$ = r regulates action type a; $tgt(r,t)$ = r has target token t; $has\_rem(f)$ = f carries `odrl:remedy`; $strong(p)$ = p is strongly permitted (profile extension, §5.3). *Activation and founding (events):* $activates(e,r)$ = e activates r [5]; *founds*/*founds_rem*/*founds_imm* = conduct / remedy / immunity founding of a relator by a rule at an event. *UFO-L relator and moments (ontology):* $Rel(\rho)$ = ρ is a UFO-L legal relator; $ODRLRel(\rho)$ = ρ is a relator founded by an ODRL rule; $bearer(m,x)$ = moment m inheres in x; $partOf(m,\rho)$ = m is part of relator ρ; $cnt(m,a,t)$ = m has normative content over action type a and target t. We inherit from UFO the *relator mediation* relation: a relator ρ mediates x and y iff it has parts inhering in x and y respectively [12, p. 240], derived from *partOf* and *bearer*, not re-axiomatised here.

Activating a permission creates a relator in which the assignee holds a Permission to Act and the assigner the correlative No Right, both over the same action and target.

Axiom 5.1 (Permission Relator Weak)**.**

$$\begin{aligned}&\forall p,x,y,a,t,e.\ \mathsf{P}(p)\wedge aee(p,x)\wedge aer(p,y)\wedge act(p,a)\wedge tgt(p,t)\wedge activates(e,p)\rightarrow\\&\quad\exists\rho_P,l,n.\ Rel(\rho_P)\wedge founds(e,\rho_P,p)\wedge\mathsf{Permission}(l)\wedge bearer(l,x)\wedge cnt(l,a,t)\\&\qquad\wedge partOf(l,\rho_P)\wedge\mathsf{NoRight}(n)\wedge bearer(n,y)\wedge cnt(n,a,t)\wedge partOf(n,\rho_P)\end{aligned}$$

A strong permission also founds a second relator at the same activation event: besides the permission relator ρ_P (Permission–No Right, Axiom 5.1), it founds the immunity relator ρ_I, which gives the assignee an Immunity and the assigner the correlative Disability.

Axiom 5.2 (Permission Relator Strong)**.**

$$\begin{aligned}&\forall p,x,y,a,t,e.\ \mathsf{P}(p)\wedge strong(p)\wedge aee(p,x)\wedge aer(p,y)\wedge act(p,a)\wedge tgt(p,t)\wedge activates(e,p)\rightarrow\\&\quad\exists\rho_I,im,db.\ Rel(\rho_I)\wedge founds\_imm(e,\rho_I,p)\wedge\mathsf{Immunity}(im)\wedge bearer(im,x)\wedge cnt(im,a,t)\wedge\\&\qquad partOf(im,\rho_I)\wedge\mathsf{Disability}(db)\wedge bearer(db,y)\wedge cnt(db,a,t)\wedge partOf(db,\rho_I)\end{aligned}$$

Activating a prohibition creates the prohibition relator ρ_F, in which the assignee owes a Duty to Omit the action a (its content is the omission $rfr(a)$, not a) and the assigner holds the correlative Right to that omission.

Axiom 5.3 (Prohibition Relator—Conduct)**.**

$$\begin{aligned}&\forall f,x,y,a,t,e.\ \mathsf{Proh}(f)\wedge aee(f,x)\wedge aer(f,y)\wedge act(f,a)\wedge tgt(f,t)\wedge activates(e,f)\rightarrow\\&\quad\exists\rho_F,d,c.\ Rel(\rho_F)\wedge founds(e,\rho_F,f)\wedge\mathsf{Duty}(d)\wedge bearer(d,x)\wedge cnt(d,rfr(a),t)\wedge\\&\qquad partOf(d,\rho_F)\wedge\mathsf{Right}(c)\wedge bearer(c,y)\wedge cnt(c,rfr(a),t)\wedge partOf(c,\rho_F)\end{aligned}$$

A prohibition with a remedy founds a competence-level relator ρ_R: the assigner gains a standing Power to declare a violation (over $decl(a)$) and the assignee the correlative Subjection.

Axiom 5.4 (Prohibition Relator—Remedy)**.**

$$\begin{aligned}\forall f,x,y,a,t,e.\ \mathsf{Proh}(f)\wedge has\_rem(f)\wedge aee(f,x)\wedge aer(f,y)\wedge act(f,a)\wedge tgt(f,t)\wedge activates(e,f)\rightarrow\\ \exists\rho_R,pw,s.\ Rel(\rho_R)\wedge founds\_rem(e,\rho_R,f)\\ \wedge\mathsf{Power}(pw)\wedge bearer(pw,y)\wedge cnt(pw,decl(a),t)\wedge partOf(pw,\rho_R)\\ \wedge\mathsf{Subj}(s)\wedge bearer(s,x)\wedge cnt(s,decl(a),t)\wedge partOf(s,\rho_R)\end{aligned}$$

Activating an obligation creates the duty relator ρ_D, in which the assignee owes a Duty to Act on a and the assigner holds the correlative Right to that action.

Axiom 5.5 (Obligation Relator)**.**

$$\begin{aligned}\forall d,x,y,a,t,e.\ obl(d)\wedge aee(d,x)\wedge aer(d,y)\wedge act(d,a)\wedge tgt(d,t)\wedge activates(e,d)\rightarrow\\ \exists\rho_D,du,c.\ Rel(\rho_D)\wedge founds(e,\rho_D,d)\wedge\mathsf{Duty}(du)\wedge bearer(du,x)\wedge cnt(du,a,t)\wedge\\ partOf(du,\rho_D)\wedge\mathsf{Right}(c)\wedge bearer(c,y)\wedge cnt(c,a,t)\wedge partOf(c,\rho_D)\end{aligned}$$

For each founding mode a rule founds one relator at a given event (and each relator has one founding event); since the modes differ, founding in two modes yields two separate relators.

Axiom 5.6 (Unique Founding)**.** For $fd\in\{founds, founds\_rem, founds\_imm\}$:

$$\forall r,\rho_1,\rho_2,e.\ fd(e,\rho_1,r)\wedge fd(e,\rho_2,r)\rightarrow\rho_1=\rho_2$$
$$\forall r,e_1,e_2,\rho.\ fd(e_1,\rho,r)\wedge fd(e_2,\rho,r)\rightarrow e_1=e_2$$

(Non-migration and temporal non-overlap: Appendix A.)

A relator is an ODRL relator when a rule of the matching kind founds it: any rule via *founds*, a prohibition via *founds_rem*, a permission via *founds_imm*.

Axiom 5.7 (ODRL Relator Typing)**.**

$$\forall e,\rho,r.\ fd(e,\rho,r)\wedge G(r)\rightarrow ODRLRel(\rho)$$

where $(fd,G)\in\{(founds,\ \mathsf{P}\vee\mathsf{Proh}\vee obl),\ (founds\_rem,\ \mathsf{Proh}),\ (founds\_imm,\ \mathsf{P})\}$.

Axiom 5.8 (ODRL Relator Is a Relator)**.**

$$\forall\rho.\ ODRLRel(\rho)\rightarrow Rel(\rho)$$

Within an ODRL relator each correlative pair comes as a matched unit: one side is present, and unique, exactly when its correlative is, so a relator never holds half a pair or a duplicate.

Axiom 5.9 (Correlativity)**.** Each ODRL simple legal relator ρ contains exactly one position from each correlative pair over the same content [8]:

$$\forall \rho, a, t.\, \mathit{ODRLRel}(\rho) \rightarrow (\exists!\, l.\, \mathsf{Permission}(l) \wedge \mathit{partOf}(l,\rho) \wedge \mathit{cnt}(l,a,t)) \leftrightarrow (\exists!\, n.\, \mathsf{NoRight}(n) \wedge \mathit{partOf}(n,\rho) \wedge \mathit{cnt}(n,a,t))$$

$$\forall \rho, a, t.\, \mathit{ODRLRel}(\rho) \rightarrow (\exists!\, d.\, \mathsf{Duty}(d) \wedge \mathit{partOf}(d,\rho) \wedge \mathit{cnt}(d,a,t)) \leftrightarrow (\exists!\, c.\, \mathsf{Right}(c) \wedge \mathit{partOf}(c,\rho) \wedge \mathit{cnt}(c,a,t))$$

$$\forall \rho, a, t.\, \mathit{ODRLRel}(\rho) \rightarrow (\exists!\, pw.\, \mathsf{Power}(pw) \wedge \mathit{partOf}(pw,\rho) \wedge \mathit{cnt}(pw,a,t)) \leftrightarrow (\exists!\, s.\, \mathsf{Subj}(s) \wedge \mathit{partOf}(s,\rho) \wedge \mathit{cnt}(s,a,t))$$

$$\forall \rho, a, t.\, \mathit{ODRLRel}(\rho) \rightarrow (\exists!\, im.\, \mathsf{Immunity}(im) \wedge \mathit{partOf}(im,\rho) \wedge \mathit{cnt}(im,a,t)) \leftrightarrow (\exists!\, db.\, \mathsf{Disability}(db) \wedge \mathit{partOf}(db,\rho) \wedge \mathit{cnt}(db,a,$$

Axiom 5.10 (Normative Position Incompatibility)**.** No agent can simultaneously bear a Permission to Act and a Duty to Omit over the same $\langle a,t \rangle$. This is a *normative* fact independent of UFO type disjointness, which governs whether a single moment individual can have two types, not whether a bearer can hold two distinct moments of incompatible types:

$$\forall l, d, x, a, t.\, \mathsf{Permission}(l) \wedge \mathit{bearer}(l,x) \wedge \mathit{cnt}(l,a,t) \wedge \mathsf{Duty}(d) \wedge \mathit{bearer}(d,x) \wedge \mathit{cnt}(d,\mathit{rfr}(a),t) \rightarrow \bot$$

Corollary 5.1 (Permission–Duty Conflict Within a Relator)**.** No ODRL simple legal relator ρ contains both a Permission to Act and a Duty to Omit over the same $\langle a,t \rangle$ for the same bearer. (Immediate from Axiom 5.10.)

Axiom 5.11 (Disability Precludes Prohibition Creation)**.**

$$\begin{aligned} \forall f, x, y, a, t.\, &\mathsf{Proh}(f) \wedge \mathit{aee}(f,x) \wedge \mathit{aer}(f,y) \wedge \mathit{act}(f,a) \wedge \mathit{tgt}(f,t) \\ &\rightarrow \neg \exists db.\, \mathsf{Disability}(db) \wedge \mathit{bearer}(db,y) \wedge \mathit{cnt}(db,a,t) \end{aligned}$$

Predicate shorthand. We write $\mathsf{Permission}(x,a,t)$ for $\exists l.\, \mathsf{Permission}(l) \wedge \mathit{bearer}(l,x) \wedge \mathit{cnt}(l,a,t)$; analogously for Duty, Right, NoRight, Power, Subj, Immunity, Disability.

Definition 5.2 (Position Instantiation)**.** For policy $\pi = \langle P,F,D \rangle$ and activation event e, the *position set* $\mathit{Pos}(\pi,e)$ is the set of legal position instances entailed by $\mathscr{A} = \{\text{Axioms 5.1–5.11}\}$:

$$\mathit{Pos}(\pi,e) = \{\, q \mid \mathscr{A} \cup \{\, \mathit{activates}(e,r) \mid r \in \pi \,\} \models q \,\}$$

The term *grounding* is reserved for the ontological interpretation of ODRL in UFO-L; $\mathit{Pos}(\pi,e)$ is the *instantiation* operation deriving concrete position individuals at a specific activation.

Proposition 5.2 (Evaluator Soundness and Incompleteness)**.** Under `closed`: evaluator permits a for x on $t \Rightarrow \mathsf{Permission}(x,a,t) \in \mathit{Pos}(\pi,e)$. Under both settings: prohibition activation $\Rightarrow \mathsf{Duty}(x,\mathit{rfr}(a),t)$; prohibition with remedy $\Rightarrow \mathsf{Power}(y,\mathit{decl}(a),t)$. The converse fails: $\mathit{Pos}(\pi,e)$ additionally entails No Right, Right to Omission, Immunity, and Disability (Table 1). The evaluator is thus *sound* but not *complete*: four of the eight entailed positions are absent from the Evaluation Report.

Example 5.1 (`:BE-Agreement` grounded)**.** Activating `:BE-Agreement` (assignee `:KulturPortal`, assigner `:BerlinerEnsemble`, action `odrl:distribute`, target

`:BE-Showtimes`) founds two relators at one event, kept distinct by Axiom 5.6. Axiom 5.3 founds the conduct relator ρ_F: `:KulturPortal` a Duty to Omit (content *rfr*(`odrl:distribute`)), `:BerlinerEnsemble` the correlative Right to Omission. As it carries a remedy, Axiom 5.4 founds the competence relator ρ_R: `:BerlinerEnsemble` a Power over *decl*(`odrl:distribute`) (to declare a violation), `:KulturPortal` the correlative Subjection. `:DT-Agreement` founds its own ρ_R for `:DeutschesTheater`, so the two theaters hold *distinct* violation-declaration Powers, the authority left implicit by the evaluator's two `violated` verdicts (Section 1).

5.3. The ODRL Legal Profile

The grounding motivates a companion OWL profile, the *ODRL Legal Profile* (`odrl-l:`)[3], enabling policy authors to assert competence-level positions at authoring time (P4). The profile defines 15 classes (1 abstract position supertype, the 8 leaf classes of Table 1, 1 abstract relator root, and 5 concrete relator classes) and 8 properties. The two ODRL extension points († in Table 1) are `odrl-l:hasRemedy` and `odrl-l:stronglyPermitted`. Three OWL approximation limits apply, all stemming from OWL's lack of function symbols and its decidable description-logic fragment: *rfr*(a) forbearances are not representable (no term for the omission as content); `stronglyPermitted` carries no OWL entailment (an annotation flag, since founding the immunity relator by Axiom 5.2 is a rule OWL cannot derive); and Axioms 5.9 and 5.10 require full FOL (a unique-existence biconditional over shared content, and a contradiction over *rfr*-related content).

6. Evaluation

We encode the axioms of Section 5.2 in three independent systems: TPTP/FOF for Vampire 5.0.0 [15] and E 3.2.5 [16], SMT-LIB 2 (`UF`) for Z3 4.15.4 [17], and Isabelle/HOL [18], which verifies all axioms and the corollary, the sanctioned-prohibition lifecycle, and Proposition 5.2. The benchmark has 39 problems over 36 identifiers (GRND001–036), three of which carry two variants for the open/closed-world, sanctioned/regimented, and with/without-immunity contrasts. Each problem is emitted as a TPTP/FOF file, an SMT-LIB 2 file, and a Turtle policy over DRK culture-dataspace entities [19], and inlines only the axioms it needs: entailment problems confirm those axioms are sufficient, discriminating problems that they are not too strong.

Consistency. The full axiom set (Ax 5.1–5.11, A.1, A.3, B1–B3) is satisfiable, witnessed by a minimal finite model (one `odrl:Permission`, two agents) from Vampire's finite-model schedule. **Entailment.** Each problem derives a grounding claim from a strict subset of axioms, including the three clauses of Proposition 5.2 and the cross-level chain behind Theorem 4.3. One instantiates the motivating scenario of Section 1: activating `drk:BerlinerEnsemble`'s prohibition over `drk:TheaterShowtimeDataset` derives exactly the Power that licenses violation declaration.

Discriminating. These witness the ontological distinctions an evaluator cannot express: open- vs. closed-world permission, sanctioned vs. regimented prohibition, weak vs. strong permission, multi-policy conflict, and the end-to-end remedy chain.

[3] `https://w3id.org/odrl-legal/`

Table 2. Axiom coverage matrix. Each row reports the SZS status of problems that exercise that axiom (possibly alongside others). $\times n$: the axiom is a schema with n FOL instances, each tested by a dedicated problem. **Thm** = theorem; **Uns** = unsatisfiable; **Sat** = satisfiable [20].

Axiom (Ref)	**SZS**	**Axiom (Ref)**	**SZS**
perm-relator-weak (Ax5.1)	Thm	correlativity ×4 (Ax5.9)	Thm/Uns
perm-relator-strong (Ax5.2)	Thm/Uns	cross-relator (Ax5.10)	Uns
proh-relator-conduct (Ax5.3)	Thm/Uns	disability-block (Ax5.11)	Uns
proh-relator-remedy (Ax5.4)	Thm	odrl-rel-is-rel (Ax5.8)	Thm
obl-relator (Ax5.5)	Thm/Sat	A1–A3, B1–B3 (§A)	Thm
unique-founding ×6 (Ax5.6)	Thm	consistency (—)	Sat
odrl-rel-typing ×3 (Ax5.7)	Thm		

All checks pass (Table 2); E omits the 3 satisfiable problems (no finite-model finder) and Z3 times out on them in the `UF` fragment, while Vampire's portfolio schedule discharges them.

7. Related Work

A formal semantics for ODRL is still emerging. The behaviour of an evaluator is described in the *ODRL Formal Semantics*, a W3C ODRL Community Group draft report (10 March 2026) [5]. The draft specifies rule applicability and the evaluation of obligations and prohibitions, providing a foundation for formal reasoning about ODRL. However, aspects such as remedy evaluation and violation handling remain underspecified and are expressed at the level of evaluator decisions rather than a formal account of authority and enforcement [5]. Steyskal and Polleres [21], and Bonatti et al. [22], provide formal and trace-based semantics for ODRL 2.1 and 2.2, capturing sanctioned interpretations and open/closed-world assumptions. Salas et al. [23,6] further develop semantic frameworks for policy comparison, query answering, and normalisation. However, these approaches do not explicitly represent the normative positions underlying evaluator decisions or competence-level authority. A survey of usage control frameworks identifies the lack of formal semantics as a persistent gap [24]. Dam et al. [25] instantiate this issue through 22 policy patterns covering achievement, maintenance, and process obligations, but without an explicit account of the normative positions or authority roles they induce. Rodríguez-Doncel and Roman [26] further show that conformance applies to policies but not processors, highlighting the lack of a unified theoretical grounding for policy enforcement. GUCON [27] provides a formal policy meta-model based on graph patterns and deontic logic, which can be instantiated using OWL 2, SHACL, and ODRL. We adopt UFO-L as the foundational ontology due to its representation of legal and social normative concepts such as commitments, claims, and powers, which align closely with deontic notions in policy languages. In contrast, foundational ontologies such as BFO [28] and DOLCE [29] provide general ontological structures but do not explicitly model normative and institutional aspects required for representing policy authority, obligations, and permissions. De Vos et al. [30] translate ODRL into Answer Set Programming using deontic fluents for permission, prohibition, and institutional power, but do not formalise the underlying normative positions between parties, as is also the case for input/output logic [31] and defeasible deontic logic [32]. Oliveira et al. [33] align ODRL with UFO-L legal moments via OWL class equivalences, but without a full axiomatic grounding of correlative normative and competence-level structures. The Soberana Ontology [34] applies UFO to model business ecosystems in data spaces, whereas our work focuses on the deontic layer for policy specification.

8. Conclusion

An ODRL evaluator returns a verdict, but not what normative situation a policy creates, who holds the authority behind it, or who may declare it violated; a formal semantics fixes the verdict, not what exists once a policy is in force. Answering that needs an ontology of legal positions rather than another account of verdicts, which is why we grounded ODRL in the foundational ontology UFO-L rather than giving it a denotational semantics. Our central result is the Cross-Level Design Principle: any policy language whose norms can be broken and carry consequences needs conduct-level positions (who may or must do what) together with competence-level positions (who may declare a violation and impose its consequences), because a conduct-only vocabulary cannot ground the step from violation to remedy. This is a necessity result, not a richer mapping, and it is what separates the work from earlier conduct-level-only alignments. Applying it to ODRL, we map each activated rule to a concrete legal relationship between assignee and assigner, surface the correlatives an evaluator leaves silent (the assigner's entitlement and the authority to declare a violation), widen what a policy makes explicit from two positions to eight, and let a companion profile state that authority at authoring time; the whole theory is checked mechanically.

Those checks rely on general-purpose provers because no ODRL engine yet covers the full vocabulary and constraint language, and the same gap makes the grounding reusable: the profile lets standard reasoners carry it on an evaluator's verdicts, querying authority, detecting conflicts, and grounding remedies without a bespoke engine. In the opening scenario the two providers' authority structures come out distinct, so which provider may demand which remedy is fixed by the model, not the implementation. The grounding covers simple legal relationships only; richer constructs such as the Liberty relator lie outside its scope. **Future Work.** (1) extend conflict detection to inheritance-based (`odrl:inheritFrom`) policies; and (2) connect the grounding to LLM-based ODRL generation, where the surfaced positions give an ontological target for policy extraction and compliance checking.

Declaration on Generative AI: The authors used Claude (Anthropic) and Grammarly for grammar and sentence refinement, reviewed all content, and take full responsibility for the publication's content.

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A. Appendix

This appendix contains full proofs of all propositions and theorems, the FOL encodings of axioms A1–A3, and the sort constraints used in the mechanized verification (Section 6).

Additional Predicates (Appendix Proofs Only)

$does(x,a,t)$ = agent x performs action a on target t; $remAct(f,b)$ = b is the remedy action of prohibition f; *NormStateChange*, *InstEvent*, *triggers*, *competentFor*, *aboutEvent* = defined in Axioms A.1–A.3 below.

Proof of Proposition 3.1

Proof. Assume regimentation: for any active prohibition on action a over asset t, performing a on t is impossible.

(1) **Normative text.** Section 2.6.2 of [1] states that a Prohibition may be infringed by the action being exercised, which triggers a requirement to fulfil all remedies. This presupposes that the prohibited action *was* performed: infringement is a reachable state.

(2) **Contradiction with assumption.** Under the assumption, performing a is impossible while the prohibition is active. The infringement condition in (1) is therefore a semantic impossibility, rendering the sentence vacuously true but describing an unreachable scenario.

(3) **Remedy construct.** Section 2.6.7 of [1] defines `odrl:remedy` as an agreed Duty that activates only upon infringement of a Prohibition by being exercised.

(4) **Remedy is vacuous under assumption.** Infringement is unreachable by (2), so `odrl:remedy` can never activate. Yet the Recommendation defines this construct, provides examples (Example 24 in [1]), and requires implementors to support it (Appendix B, criterion 9 [1]).

(5) **Contradiction.** The Recommendation treats `odrl:remedy` as a meaningful implementable construct; the regimentation assumption renders it semantically vacuous. These cannot both hold. Assumption is false.

ODRL prohibition is not regimented. The Formal Semantics [5] confirms this independently: it defines an explicit `violated` state reached when the regulated action has been performed [5]. Since prohibitions are violable and violation triggers normative consequences, they are sanctioned in the sense of Jones and Sergot [35]. □

Proof of Proposition 3.2

Proof. The ODRL Formal Semantics defines three evaluator states for obligations: `fulfilled` upon a single performance event, `violated` when performance is no longer possible, and `not-set` otherwise [5, §6]. Each transition is triggered by a bounded performance event or its absence, presupposing an *achievement* satisfier in the sense of [13,14].

A *maintenance* obligation is satisfied by a persisting state, not a bounded event: the evaluator has no transition for "state holds continuously". A *process* obligation is satisfied by an ongoing activity: since being ongoing is a temporary property incompatible with the timeless character of events [14], the three-state architecture cannot represent process satisfaction. Neither maintenance nor process obligations have a defined `fulfilled` or `violated` transition under the current semantics. The proposition follows. □

Proof of Proposition 4.1

Proof. Weak permission is absence of obligation to refrain [36]. The pair

$$\{\mathsf{Permission}(x,a,t),\, \mathsf{NoRight}(y,a,t)\}$$

captures this fully: the assignee holds a Permission to Act, the assigner holds the correlative No Right. Since weak permission makes no persistence claim, no model extension requires additional positions: the pair remains adequate under any extension consistent with **A1**–**A3**. No Power, Immunity, or Disability is needed. □

FOL Encoding of Axioms A1–A3

The following first-order encodings underlie the mechanized verification in Section 6. The prose statements appear in Section 4. All three axioms are imported into the TPTP/-FOF and SMT-LIB encodings described in Section 6.

Axiom A.1 (A1: Normative State Changes Require an Institutional Event)**.**

$$\forall x,a,t,q.\; \mathit{NormStateChange}(x,a,t,q) \rightarrow \exists e.\; \mathit{InstEvent}(e) \wedge \mathit{triggers}(e,x,a,t,q)$$

A normative position q coming into or going out of existence for agent x over action a on target t requires a triggering institutional event e.

Axiom A.2 (A2: Institutional Events Require a Competent Agent)**.**

$$\forall e.\; \mathit{InstEvent}(e) \rightarrow \exists y.\; \mathit{Agent}(y) \wedge \mathit{competentFor}(y,e)$$

No institutional event occurs without an agent holding the relevant competence to bring it about.

Axiom A.3 (A3: Competence Is a Power–Subjection Pair)**.**

$$\begin{aligned}
\forall y,e.\; &\mathit{competentFor}(y,e) \rightarrow \exists pw,s,x.\\
&\quad \mathsf{Power}(pw) \wedge \mathit{bearer}(pw,y) \wedge\\
&\quad \mathsf{Subj}(s) \wedge \mathit{bearer}(s,x) \wedge\\
&\quad \mathit{aboutEvent}(pw,e) \wedge \mathit{aboutEvent}(s,e)
\end{aligned}$$

Competence to perform institutional event e is grounded in a Power–Subjection pair: y holds the Power, some x the correlative Subjection.

New predicates. $NormStateChange(x,a,t,q)$: position q over $\langle a,t\rangle$ changes for x; $InstEvent(e)$: e is an institutional event; $triggers(e,x,a,t,q)$: e triggers the change of q; $competentFor(y,e)$: y is competent to perform e; $aboutEvent(pw,e)$: Power/Subjection pw concerns event e; $Agent(y)$: y is an agent capable of performing institutional acts (subsumes assignees and assigners).

Bridge axioms (connect A1–A3 to main-body predicates). B1–B3 are declared in the TPTP/SMT-LIB signatures alongside A1–A3. They do not appear in the main-body axioms, which operate only on the derived shorthand positions.

B1 (violation is a normative state change):

$$\begin{aligned}&\forall f,x,a,t.\ \mathsf{Proh}(f)\wedge has\_rem(f)\wedge act(f,a)\wedge tgt(f,t)\wedge aee(f,x)\wedge does(x,a,t)\rightarrow\\ &\qquad\exists b.\ remAct(f,b)\wedge NormStateChange(x,b,t,Duty_{rem})\end{aligned}$$

where $remAct(f,b)$ holds iff b is the action of the remedy rule attached to f. Performing a in violation of f constitutes a normative state change activating precisely the reparative Duty over b.

B2 (Power content links to founding event):

$$\begin{aligned}\forall pw,a,t,\rho,e,f.\ \mathsf{Power}(pw)\wedge cnt(pw,decl(a),t)\wedge&\\ partOf(pw,\rho)\wedge founds\_rem(e,\rho,f)\rightarrow&\\ aboutEvent(pw,e)&\end{aligned}$$

B3 (Subjection content links to founding event):

$$\begin{aligned}\forall s,a,t,\rho,e,f.\ \mathsf{Subj}(s)\wedge cnt(s,decl(a),t)\wedge&\\ partOf(s,\rho)\wedge founds\_rem(e,\rho,f)\rightarrow&\\ aboutEvent(s,e)&\end{aligned}$$

Proof of Proposition 5.1 (Open-World Normative Vacuum)

Proof. No axiom in $\mathscr{A}$ fires on an unregulated action: Axioms 5.1–5.11 all require $activates(e,r)$ for some rule $r\in\pi$ regulating a. If no such r exists, no existential is triggered and $Pos(\pi,e)$ contains no position over $\langle a,t\rangle$. The evaluator's open-world default is therefore a runtime convention with no counterpart in $\mathscr{A}$. □

ODRL Legal Profile

The full profile is available in the benchmark repository.[4]

[4] `https://github.com/Daham-Mustaf/odrl-ufol-grounding/blob/master/ODRL-Legal-Profile/odrl-legal-profile.ttl`

Proof of Theorem 4.1

Proof. **Part 1: Conduct-only characterization is inadequate.** Let $\mathscr{H}_1 = \{\mathsf{Permission}(x,a,t), \mathsf{NoRight}(y,a,t)\}$. Consider a model $\mathscr{M}$ satisfying $\mathscr{H}_1$. Extend $\mathscr{M}$ to $\mathscr{M}'$ by adding a prohibition issued by y with assignee x over action a on target t. By Axiom 5.3, $\mathscr{M}'$ contains $\mathsf{Duty}(x, \mathit{rfr}(a), t)$.

No agent can simultaneously bear a Permission to Act and a Duty to Omit over the same $\langle a,t\rangle$: by Axiom 5.10 (Normative Position Incompatibility), $\mathsf{Permission}(x,a,t)$ and $\mathsf{Duty}(x, \mathit{rfr}(a), t)$ cannot be co-borne—this is a *normative* fact independent of UFO type disjointness, which governs whether a single moment individual can have two types, not whether a bearer can hold two distinct moments of incompatible types [8]. $\mathscr{M}'$ is therefore inconsistent. Strong permission is intended to *persist* under model extension. Since $\mathscr{H}_1$ does not preserve this, it is inadequate.

Part 2: Cross-level characterization is adequate.
Let $\mathscr{H}_2 = \mathscr{H}_1 \cup \{\mathsf{Immunity}(x,a,t), \mathsf{Disability}(y,a,t)\}$. By Axiom 5.11, $\mathsf{Disability}(y,a,t)$ entails that no prohibition issued by y over $\langle a,t\rangle$ can exist. Therefore under any model extension $\mathscr{M}'$, y cannot create $\mathsf{Duty}(x, \mathit{rfr}(a), t)$, and x's $\mathsf{Permission}(x,a,t)$ is preserved. $\mathscr{H}_2$ is adequate.

Collective assigners. When y is a collective agent, Disability distributes to each member y_i, ensuring Theorem 4.1 holds in multi-assigner scenarios [8].

Conclusion. $\mathscr{H}_1$ is inadequate; $\mathscr{H}_2$ is adequate. Since Immunity and Disability are competence-level positions (§2.2), strong permission is cross-level under **A1**–**A3**. □

Proof of Theorem 4.2

Proof. Sanctioned prohibition requires three conditions: (i) $\mathsf{Duty}(x, \mathit{rfr}(a), t)$ (duty to refrain), (ii) the action a is performable despite the duty (violability), and (iii) violation triggers normative consequences (a reparative obligation becomes active).

Part 1: Conduct-only is inadequate. Let $\mathscr{H}_3 = \{\mathsf{Duty}(x, \mathit{rfr}(a), t), \mathsf{Right}(y, \mathit{rfr}(a), t)\}$. This captures (i) and (ii). For (iii): when x performs a, Bridge axiom B1 yields $\mathit{NormStateChange}(x, b, t, \mathit{Duty}_{rem})$. By **A1**, this requires an institutional event e'. By **A2**, e' requires a competent agent. By **A3**, that competence is a Power–Subjection pair. None of these exist under $\mathscr{H}_3$, so the violation-to-remedy transition has no ontological grounding. $\mathscr{H}_3$ is inadequate.

Part 2: Cross-level is adequate. Let $\mathscr{H}_4 = \mathscr{H}_3 \cup \{\mathsf{Power}(y, \mathit{decl}(a), t), \mathsf{Subj}(x, \mathit{decl}(a), t)\}$. By Axiom 5.4, these positions are bundled in a distinct competence relator ρ_R founded by $\mathit{founds\_rem}(e, \rho_R, f)$ at the same activation event e that founds ρ_F. Bridge axiom B2 yields $\mathit{aboutEvent}(pw, e)$. By **A3**, y's competence is grounded in pw. By **A2**, y can perform the institutional event $e' = \mathit{decl}(a)$. By **A1**, e' triggers *NormStateChange*, activating the remedy obligation. $\mathscr{H}_4$ grounds the full lifecycle: duty $\rightarrow$ violation $\rightarrow$ declaration $\rightarrow$ remedy active. $\mathscr{H}_4$ is adequate.

Conclusion. $\mathscr{H}_3$ is inadequate; $\mathscr{H}_4$ is adequate. Since Power and Subjection are competence-level, sanctioned prohibition is cross-level under **A1**–**A3**. □

Proof of Theorem 4.3

Proof. Let N be any norm that is (i) violable and (ii) consequential. By (ii), violation of N activates a reparative normative position—a *NormStateChange* in the sense of **A1**. By **A1**, this requires a triggering institutional event. By **A2**, that event requires a competent agent. By **A3**, that competence is a Power–Subjection pair, which are competence-level positions. Hence no characterization using only conduct-level positions is adequate: it cannot ground the violation-to-consequence transition. N is therefore cross-level.
Regimented norms (violation impossible) satisfy condition (ii) vacuously and require no competence-level positions. Inconsequential norms satisfy neither condition and are conduct-level-complete by Proposition 4.1.
Since ODRL prohibition is sanctioned (Proposition 3.1) and remedies are normative consequences of violation, ODRL prohibition satisfies both (i) and (ii), and the theorem applies. □

Proof of Principle 1

Proof. Immediate from Theorem 4.3. That theorem establishes: for any norm satisfying (i) violable and (ii) consequential, no conduct-only set of positions is an adequate characterization. Principle 1 is the metalinguistic corollary: a language whose normative vocabulary is restricted to conduct-level positions cannot express an adequate characterization of any such norm, hence cannot serve as a complete grounding for any normative system containing violable, consequential norms. The language must therefore include Power, Subjection, Immunity, and Disability alongside Permission, Duty, Right, and No right. Regimented norms are excluded by condition (i); inconsequential norms by condition (ii); both are conduct-level-complete by Proposition 4.1. □

Proof of Proposition 5.2 (Evaluator Soundness and Incompleteness)

Proof. **Soundness.** Each evaluator verdict traces to one axiom: permission active implies $activates(e,p)$, so Ax5.1 entails $\mathsf{Permission}(x,a,t)$; prohibition active implies $activates(e,f)$, so Ax5.3 entails $\mathsf{Duty}(x,rfr(a),t)$; prohibition with remedy implies $has\_rem(f)$, so Ax5.4 entails $\mathsf{Power}(y,decl(a),t)$.

Incompleteness. The W3C Evaluation Report [5, §2] defines no output field for No Right, Right to Omission, Immunity, or Disability. Each is entailed by the axioms: No Right by Ax5.1 (correlative of Permission); Right to Omission by Ax5.3 (correlative of Duty to Omit); Immunity and Disability by Ax5.2 (when $strong(p)$). The evaluator is therefore sound but not complete with respect to $Pos(\pi,e)$. □

Notes on Axiom 5.6 (Unique Founding)

Non-migration and temporal non-overlap. Policy revision destroys the old relator and founds a fresh one with a historical dependence relation to the old [10,12]; two activations of the same rule over identical relata must be temporally non-overlapping [10].

Normative Content Sort Constraints

These constraints are referenced in Section 5.2. They are not ontological axioms but signature constraints: they make the single-sorted FOF/SMT-LIB encoding faithfully represent the two-sorted content domain that UFO-L's moment typing presupposes. In Isabelle/HOL they are absorbed by the type system and do not appear as axioms.
Here *pos* is the left-inverse of *rfr*: $pos(rfr(a)) = a$ recovers the underlying act from its omission, used by provers to discharge injectivity goals without quantifier alternation.

$$\forall x.\, Act(x) \rightarrow \neg Forbearance(x)$$

$$\forall a.\, Act(a) \rightarrow Forbearance(rfr(a))$$

$$\forall a.\, rfr(a) \neq a$$

$$\forall a, b.\, rfr(a) = rfr(b) \rightarrow a = b$$

$$\forall a.\, pos(rfr(a)) = a$$

$$\forall a.\, Act(a) \rightarrow Act(decl(a))$$